ICT303 – Advanced Machine Learning and Artificial Intelligence

Topic 1: Introduction and Admin Matters

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How to Get in Touch with the Teaching Team

- Internal and External Students
  - Email: H.Laga@murdoch.edu.au.
- Important
  - In any communication, please make sure that you
    - Start the subject of your email with ICT303
    - Include your student ID, name, and the lab slot in which you are enrolled.
  - We will do all our best to answer your queries within 24 hrs.

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In this Lecture

- Introduction to Advanced Machine Learning (ML) and AI
  - Learning objectives
  - What is covered and what is not
- Admin matters
  - Unit Overview
  - Assessment Components
  - How to pass the unit
- Summary

- Learning objectives
  - Have a general idea of what is Machine Learning and Deep Learning
  - Understand the scope of the unit
  - Understand the expectations of the unit and the different assessment components
- Additional readings
  - Chapter 1 of the textbook, available at: https://d2l.ai/

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What is Machine Learning (ML)

- Traditional computer programs and programming
  - A list of instructions that, when executed in a logical sequence, solve a specific problem
  - Their performance does not change with experience – they always produce the same output when presented with the same input
  - The performance will only change if the developers change the code

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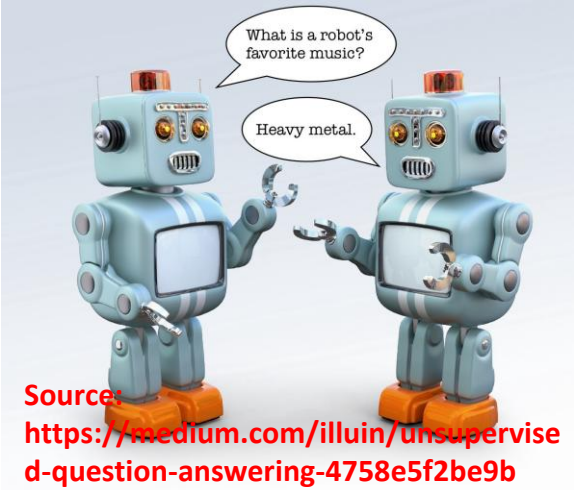
What is Machine Learning (ML)

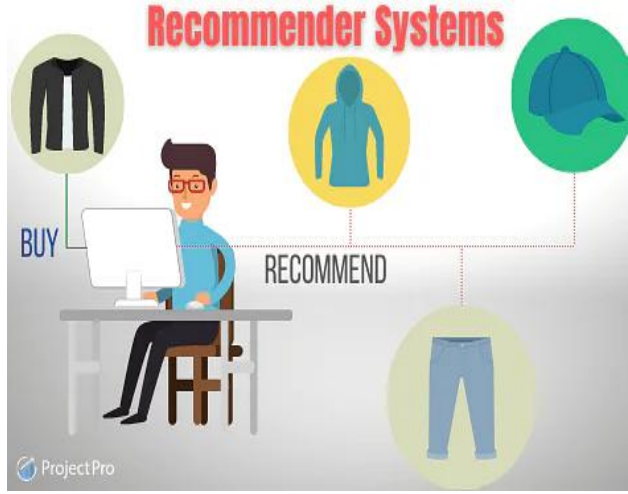
- Can you write a list of code instructions that will automatically:

Count how many people in an image?

Describe what people are doing in this scene?



Converse, understand and answer questions in a natural language?



Make shopping recommendations to a user?

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Simpler than that ….

- Write a program that when a user says: “Hey Siri”, it will wake up the Siri app on your iPhone
  - Input:
    - The sound wave of the user’s voice when pronouncing the sentence
  - Output:
    - Yes (or 1), means your program has recognized the sentence “Hey Siri”
    - No (or 0), means the sentence pronounced by the user is different
  - The program should be a list of instructions that will explicitly tell the computer how to map a sound wave into {Yes, No}

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What is Machine Learning (ML)

- Traditional programs cannot solve many problems that look simple but difficult to describe as a list of instructions
  - Sometimes the program that we are looking for
    - Does not follow any (clear) pattern
    - Follows a pattern that changes over time; thus, the program needs to be continuously adapted
  - The relation between the parameters (e.g., parameters that affect weather, share market historical data) and the output (weather, share price) may be unclear/unknown or too complicated
    - Thousands / millions of computations would be required to model this type of relationships

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What is Machine Learning (ML)

- Machine Learning (ML) is the study of techniques that can learn from experience
  - Experience can be observational data, historical data, interactions with an environment,
  - The more experience a ML algorithm accumulates, the more its performance improves
- Example: "Hey Siri" program
  - Do not attempt to design a deterministic program (with an explicit list of instructions) to recognize "Hey Siri"
  - Design a flexible program (called model) whose behaviour is determined by a set of parameters
  - Use a training dataset to determine/learn the best possible set of parameters, those that provide the best performance
    - Performance is measured with respect to some objective (or loss) function

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Key Components of Machine Learning

- Data
  - The data we can learn from
- A model
  - A model of how to transform (map) the data into the desired output
- Training algorithm, which is in fact an optimization algorithm
  - An algorithm to adjust the model's parameters to optimize the objective function, i.e., to obtain the best results when the model is given as input the training data
- An objective function
  - A function that quantifies how well (or badly) the model is doing

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Key Components of Machine Learning - Data

- Cannot do Machine Learning (as well as data science) without DATA
  - A collection of examples from which the ML model must learn its predictions
  - It should contain examples of input and their corresponding (desired) output
    - The desired output for each input is called ground-truth label or just label
    - Sometimes the labels are called annotations

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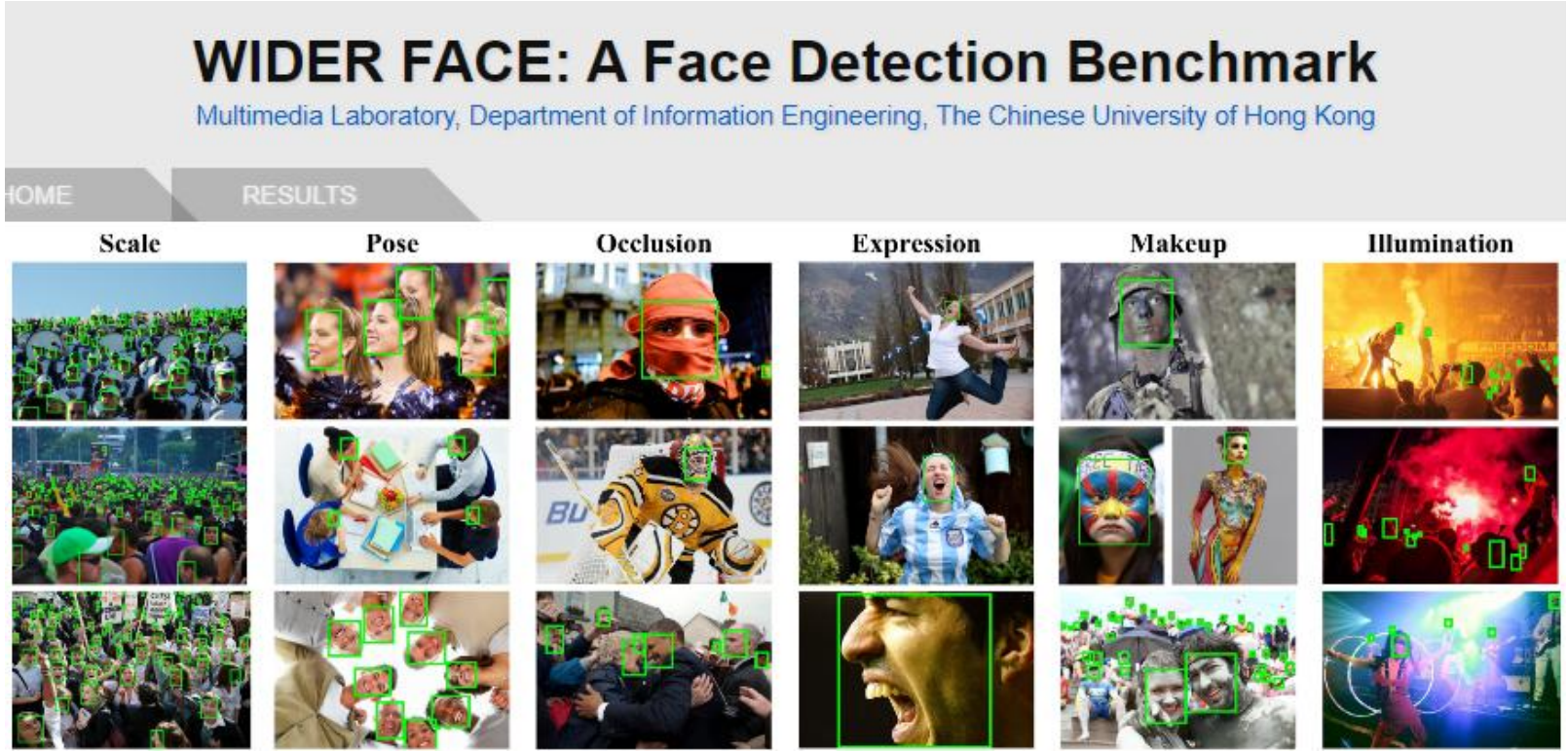
Key Components of Machine Learning - Data

- Example 1 – Face detection
  - Images captured in various conditions
  - Labels are the location (in the form of bounding boxes) of faces in each image



http://shuoyang1213.me/WIDERFACE/

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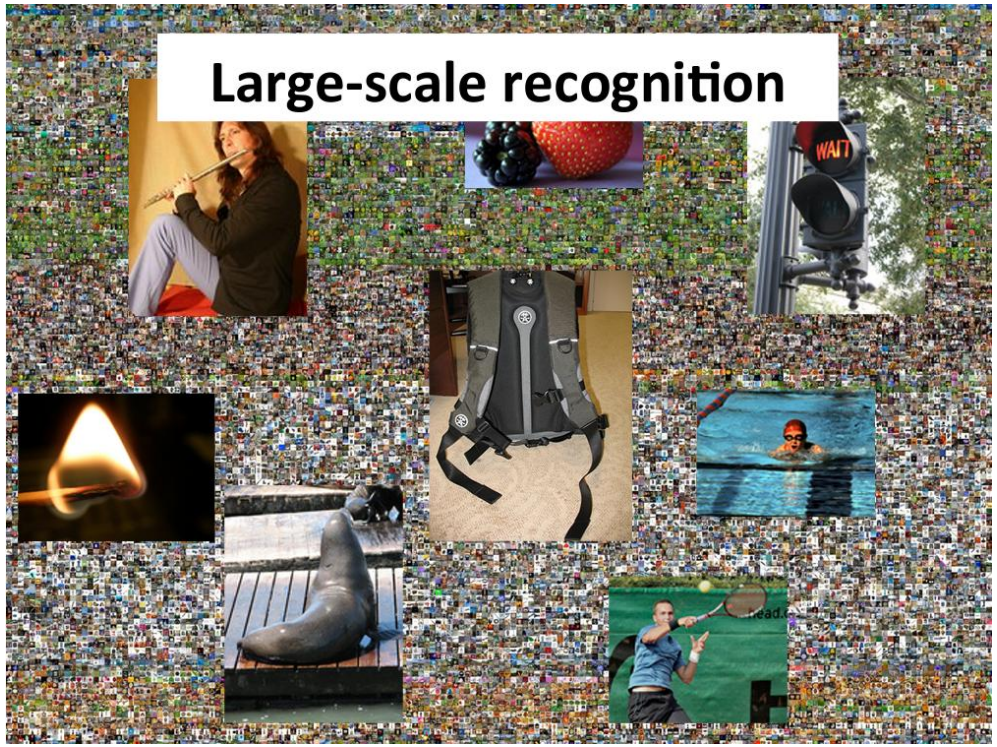
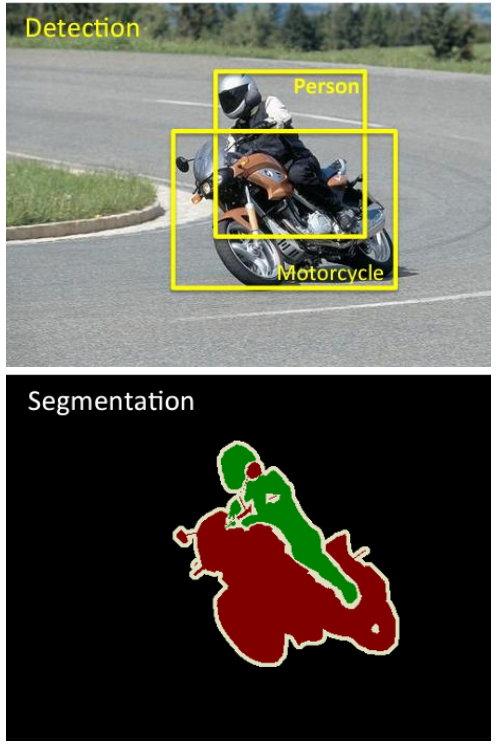
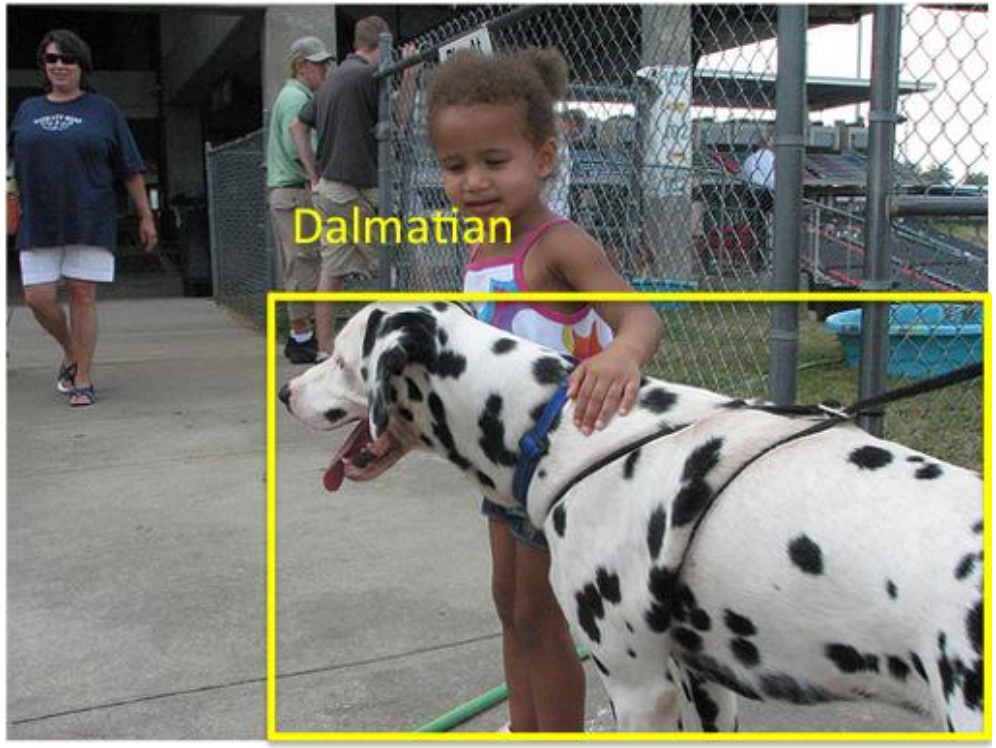
Key Components of Machine Learning - Data

- Example 2 – Object recognition

1000 object classes **1,431,167 images**







ImageNet dataset: https://www.image-net.org/

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Key Components of Machine Learning - Data

- Cannot do Machine Learning (as well as data science) without DATA
  - A collection of examples from which the ML model must learn its predictions
- In general, the more data we have the better
  - Quantity, however, is <u>not</u> everything
  - We need to have the right data with correct ground truth!
    - If you feed the model garbage data, you will get garbage predictions (garbage in, garbage out)
    - It should be free (as much as possible) of mistakes
  - Ideally, the data should cover all (or most of) the possible cases,
    - e.g., in object recognition, all classes of objects should have more or less equal no. of instances in the training datasets

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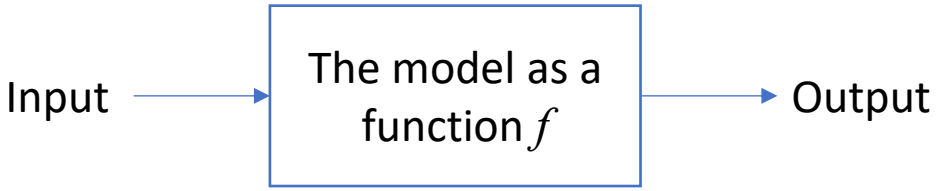
Key Components of Machine Learning – The Model

- A model
  - A function that transforms (maps) the data into the desired output
  - The function can be
    - Simple, thus can have a closed analytical form
    - Complex – we cannot write it down in the form of a mathematical formula
  - The function is controlled by some parameters



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Key Components of Machine Learning – The Model

- Examples of models

Simple Moving Average for stock price prediction

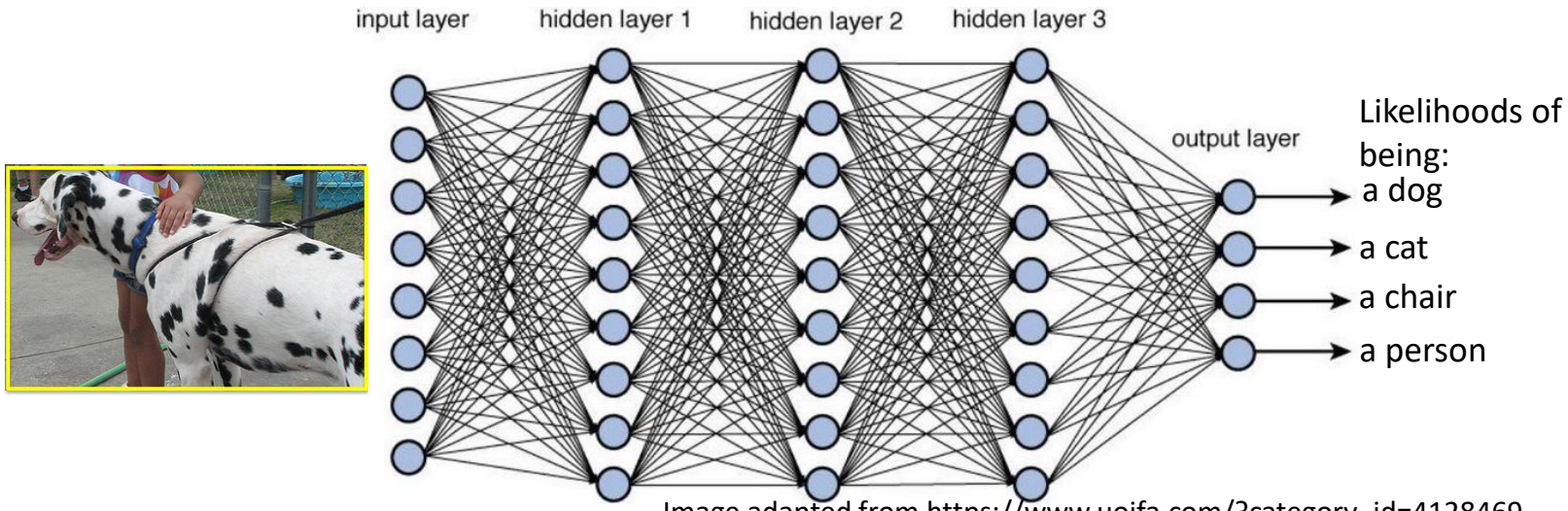

Image adapted from https://www.uoifa.com/?category_id=4128469

A Neural Network

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Key Components of Machine Learning – The Model

- Examples of models - Deep Neural Networks
  - They have millions of parameters

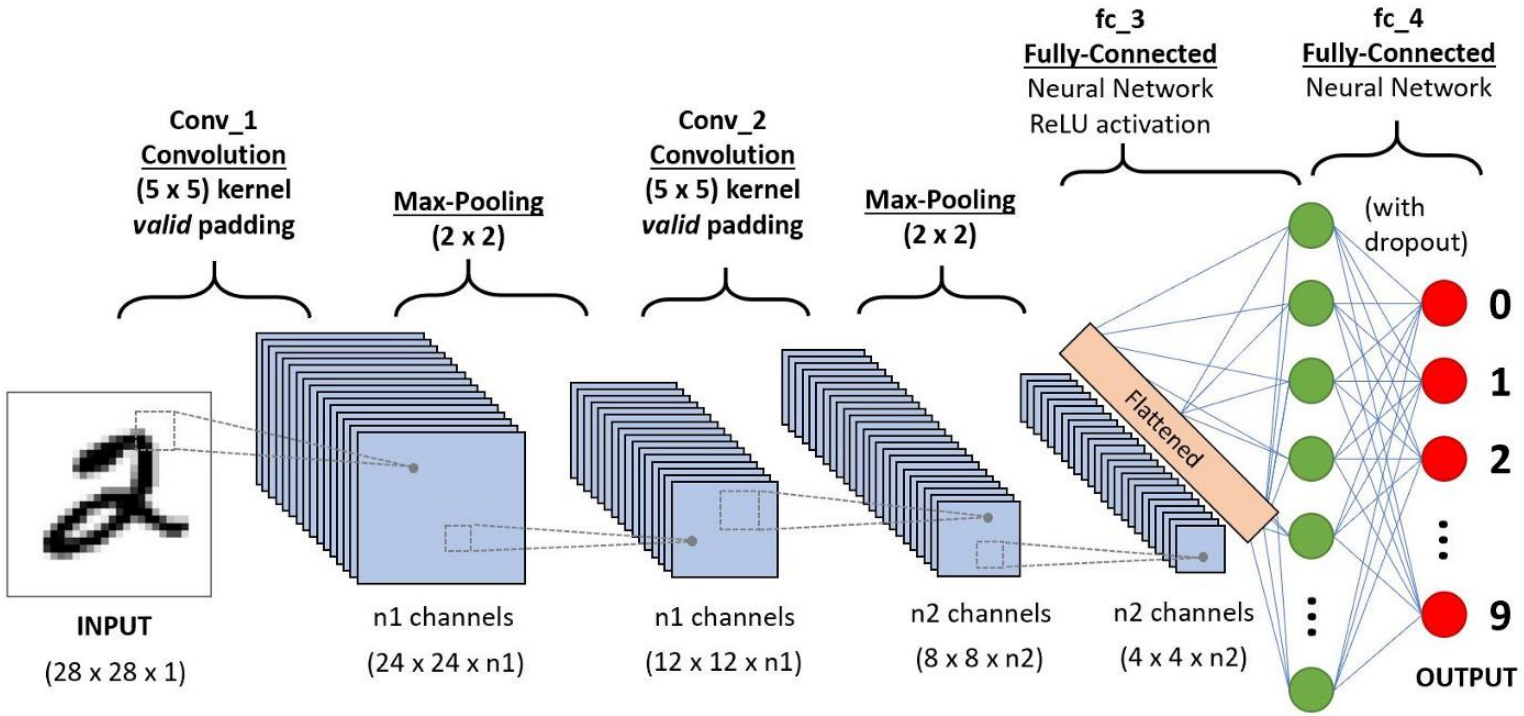

Image courtesy of: https://towardsdatascience.com/a-comprehensive-guide-to-convolutional-neural-networks-the-eli5-way-3bd2b1164a53

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Key Components of Machine Learning – Training

- Once we have got training data and a model, we need an algorithm capable of searching for the best possible parameters so that the model produces the correct output for given input
  - The model will be presented with an input
  - It will compute an output
  - Measure the discrepancy (difference/error) the computed output and the desired (correct) output
  - Adjust its parameters so that this error is reduced, then repeat the steps above
- Thus, it needs
  - A function, called objective / loss function, to measure the error
  - An optimization algorithm to minimize this objective function
- Popular optimization algorithms (will cover them later) are
  - Gradient descent, Stochastic gradient descent, etc.

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Key Components of Machine Learning – Objective Functions

- Learning is the process of improving at that task over time
  - How to measure improvement?
    - We need formal measures of how good (or bad) the machine learning models are
  - These measures are called *Objective Functions* or *Loss Functions*
    - Think of it as a measure of error in the prediction - The lower the objective function is the better
    - It is defined with respect to the model's parameters and depends upon the dataset
  - Training is the process of finding the model's parameters so that the loss function is minimized

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Learning Process

- Learning
  - It is the process by which we discover the right setting (values) of the parameters which will result in the desired behaviour of the model

- Learning process
  - Step 1
    - Start with random values for the parameters of the model
  - Step 2
    - Grab some of the (training) data along side with the corresponding desired output (called labels)
  - Step 3
    - Tweak the model parameters so that the model performs slightly better
  - Step 4
    - Repeat Steps 2 and 3 until the model is awesome

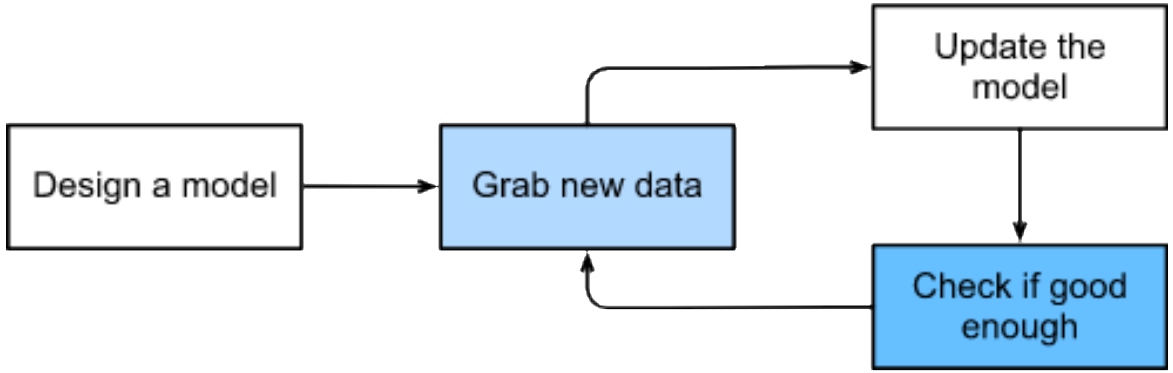

Image courtesy of https://d2l.ai/

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Types of Machine Learning

- Supervised Learning
- Unsupervised and Self-Supervised Learning
- Reinforcement Learning
- Few and zero-shot learning

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Types of Machine Learning - Supervised Learning

- During training, the model is given examples composed of an input (e.g., image of an object) and the desired output, i.e., ground truth label (e.g., dog)
  - The input goes through the model
  - The model then returns the prediction (e.g., the "estimated" label)
  - The loss function then computes the error (discrepancy) between the ground-truth and the prediction
  - The optimization algorithm then updates the parameters of the model so that in the next iteration, this error is reduced
  - Repeat the process until we are satisfied that the error is low

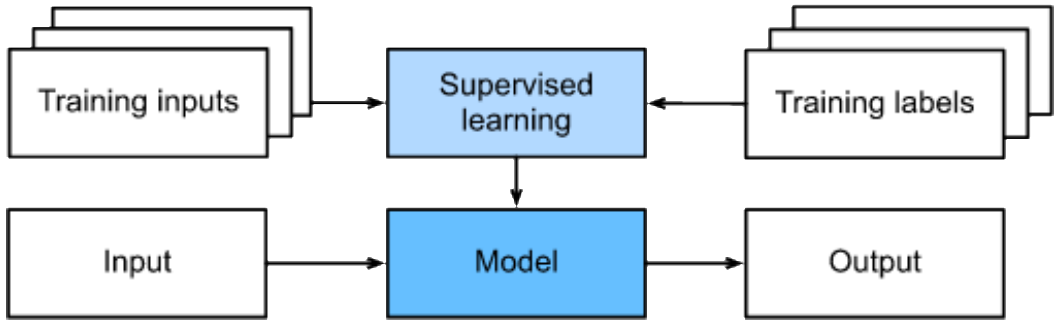

Image courtesy of https://d2l.ai/

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Types of Machine Learning - Unsupervised / Self-supervised

- Unsupervised / self- supervised is the case where the ground-truth labels are not available. Example of clustering
  - Given a set of images, can we group them into meaningful categories (we do not know in advance what are these categories)
  - Given a collection of internet browsing activities of various users, can we cluster users into profiles?

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Types of Machine Learning - Reinforcement Learning

- Used to develop an agent that interacts with the environment & takes actions
- During training
  - The agent interacts with the environment over a series of time steps
  - At each time step,
    - The agent receives some observation from the environment
    - Must choose an action that is transmitted back to the environment
    - The agent receives a reward from the environment
  - The goal of the training is to maximize the reward

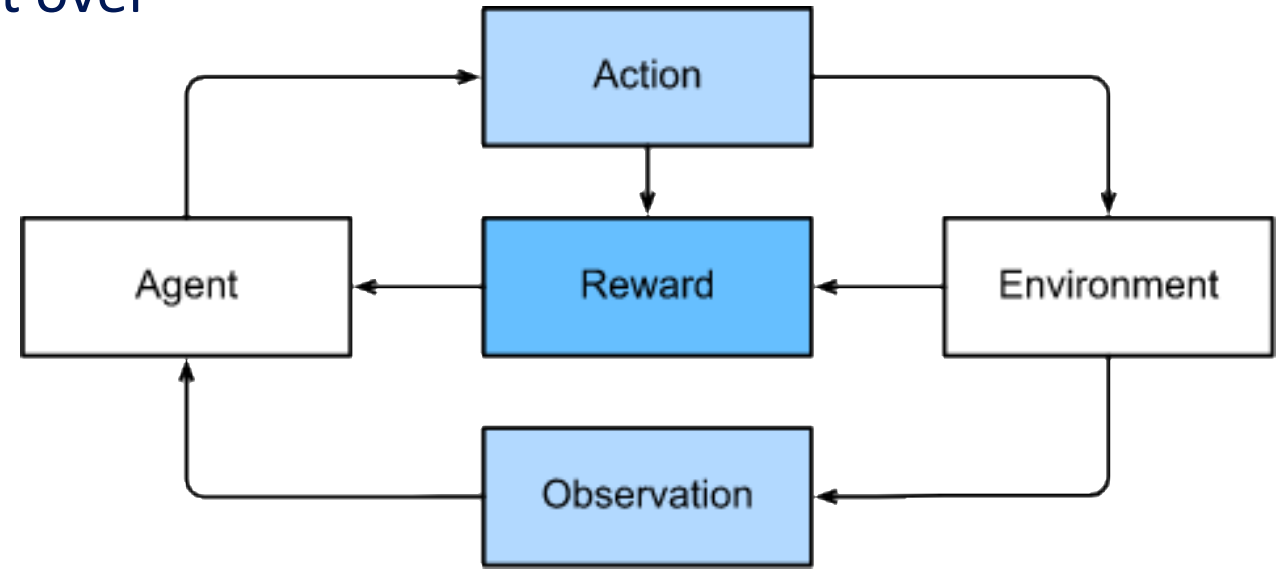

Image courtesy of https://d2l.ai/

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Types of Machine Learning – Few-Shot Learning

- Few shot learning (FSL), or low-shot learning (LSL), is the type of machine learning methods where the training dataset contains limited information
- Why is it important?
  - Learning like humans – e.g., humans can spot the difference between handwritten characters after just seeing a few examples
  - Learning for rare cases – e.g., classifying rare species of animals
  - Reducing data collection effort
- One-shot learning is a special case of few-shot learning

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Types of Machine Learning – Zero Shot Learning

- Zero-shot learning aims to predict the correct output (e.g., the type of animals in an image) without being exposed to any instances belonging to that class in the training stage
  - The model will be trained on some classes
  - It should be able to generalize to a new category (class) of samples although it hasn't seen it during the training
  - Example
    - A model trained to distinguish between cats and dogs is made to identify images of birds
- Why is it important?
  - Learning for rare cases – e.g., classifying rare species of animals, diseases, etc. where it is difficult to get training data
  - Reducing data collection effort

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The Roots of Machine Learning

- The core ideas behind learning from data has been studied for centuries
  - Jacob Kobel (1460 – 1533) – averaging the length of 16 adult men's feet to estimate the average foot length in the population
  - Bernoulli distribution (1655 – 1705) and Gaussian distribution (1777 – 1855)
  - Gauss (1777 – 1855) invented least mean squares algorithm

- Read Section 1.4 of https://d2l.ai/chapter_introduction/index.html for a few interesting historical facts

https://d2l.ai/chapter_introduction/index.html

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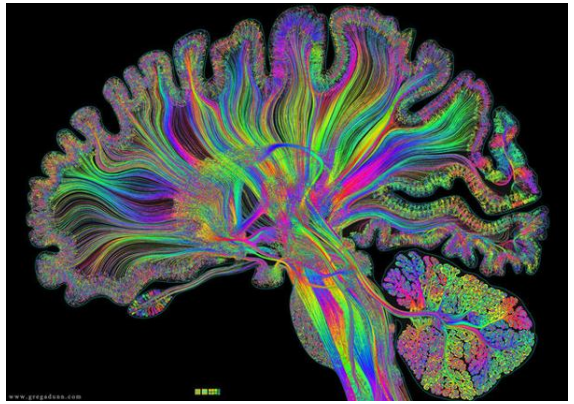
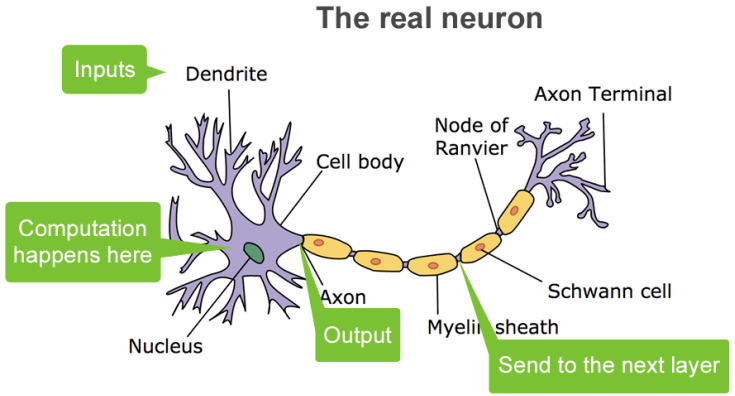
The Road to Deep Learning

- Artificial Neurons (1944)
  - Neurophysiologist Warren McCulloch and mathematician Walter Pitts wrote a paper on how neurons might work
    - They modeled a simple neural network using electrical circuits.
  - A lot of money has been then put to fund such research

- 1975 - 1986
  - 1975 – first multilayered network
  - 1982 – John Hopfield created bidirectional connections between neurons (previously, connections were unidirectional)
  - 1986 – multiple layered neural networks and backpropagation algorithm to train them
    - NN at that time did not achieve the performance people were expecting (issues with computation and training data)
  - Quickly abandoned in favor of Bayesian Machine Learning



https://c.d2l.ai/berkeley-stat-157/syllabus.html



https://cs.stanford.edu/people/eroberts/courses/soco/projects/neural-networks/History/history1.html

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The Road to Deep Learning

- The modern era – Three important factors changed the world
  - The fundamentals of NN, especially the Multilayer perceptron and backpropagation algorithms (1980s)
  - The internet, especially Google, Facebook and Amazon (and others)
    - People are sharing all sorts of data (and even labelling them) for free and at large scale
    - A lot of data are being automatically collected without us knowing (and often without our consent)
  - High Performance Computing
    - GP-GPUs (Nvidia's GPUs in particular) enabled massive parallel computing on desktops
- Major companies (as well as universities) started developing and training large neural networks on massive data
  - And thus the birth of Deep Learning

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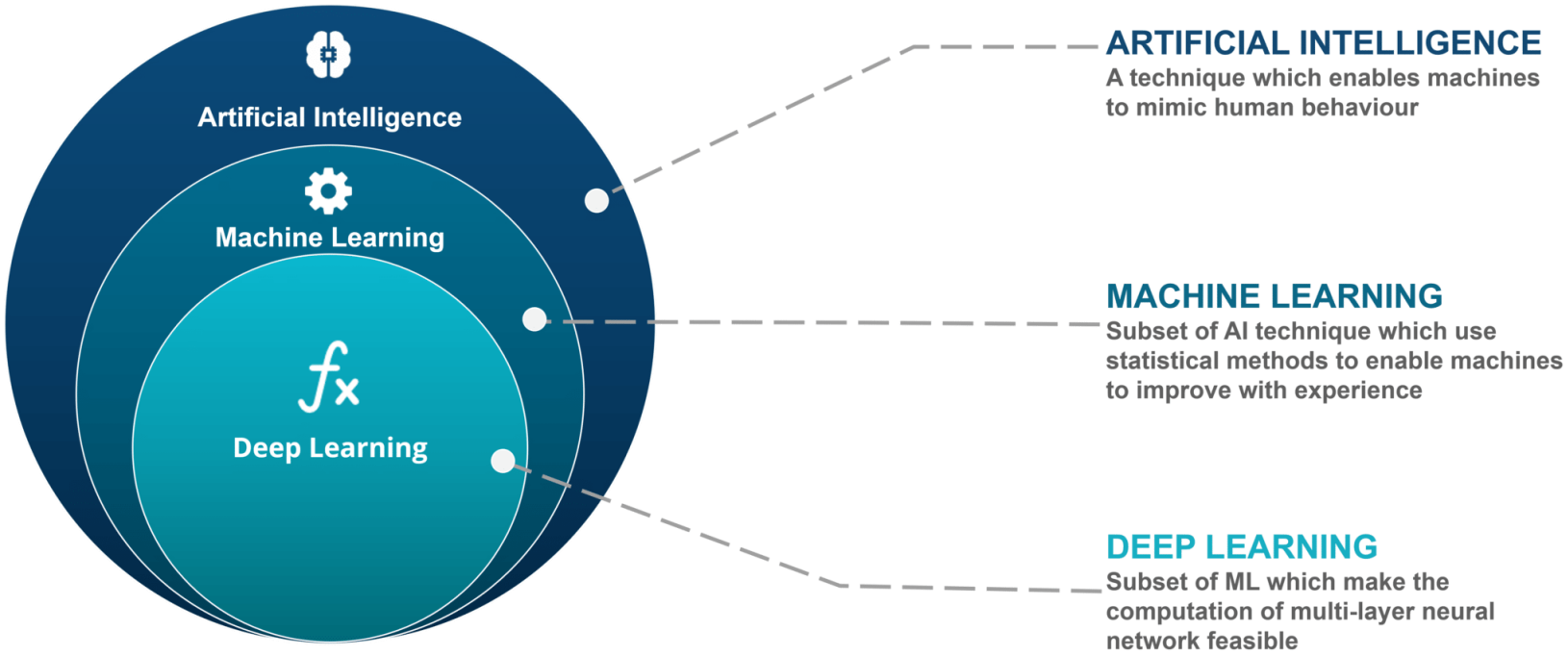
Deep Neural Networks and Deep Learning

- Deep learning (DL) is a subset of machine learning concerned with models based on many layered neural networks
  - Deep because it has many layers of (nonlinear) transformations



https://www.edureka.co/blog/ai-vs-machine-learning-vs-deep-learning/

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What about Generative AI?

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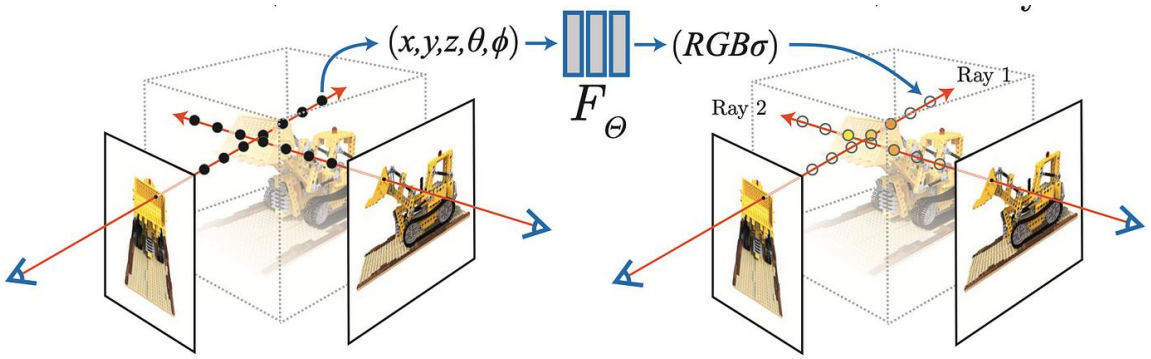
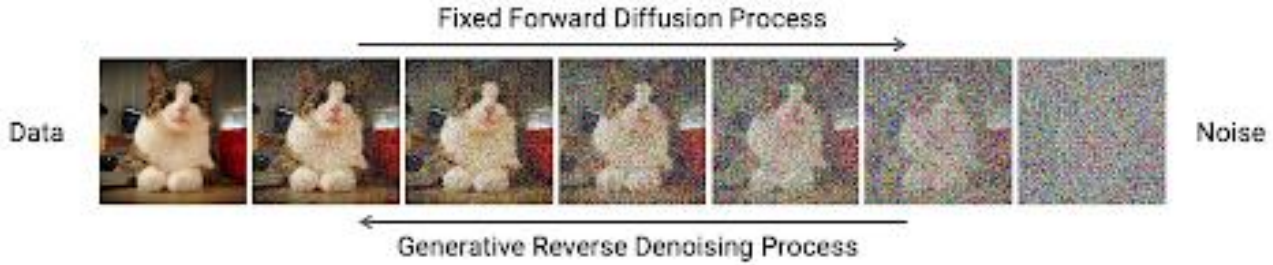
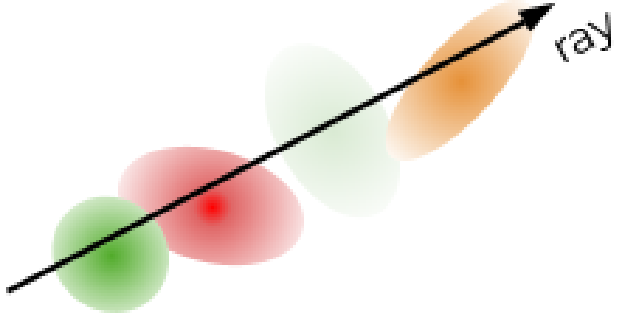
The 4 Breakthroughs (that enabled Generative AI)



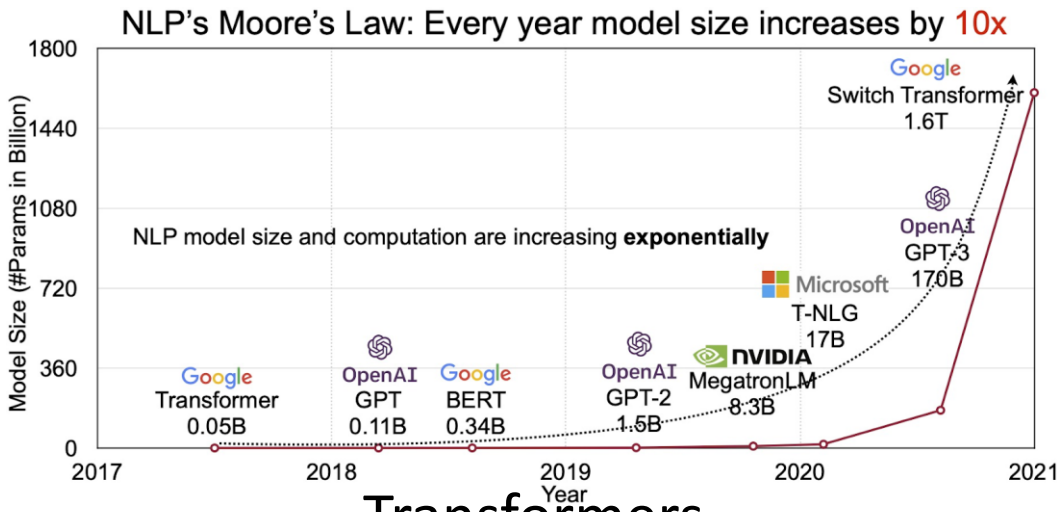
Neural Radiance Fields (NeRF)



Diffusion models



Gaussian Splatting



Transformers

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In this Unit

- We will focus on Deep Learning (DL) and Deep Neural Networks (DNN) - Theory and applications (hands-on)
  - ULO1:
    - Understand the theory and algorithms behind machine learning models
  - ULO2:
    - Develop and implement machine learning models using programming languages.
  - ULO3:
    - Train, test and evaluate the performance of machine learning models.
  - ULO4:
    - Application of best practices for data preparation.
  - ULO4:
    - Use deep learning models to solve practical problems in various domains.
  - ULO5:
    - Learn new programming and scripting languages without supervision.

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In this Unit

- Topics
  - Linear NN
  - Multilayer Perceptron
  - Training
  - Convolutional Neural Networks (CNN)
  - Modern CNNs (ResNet, Recurrent NN, Transformers, etc.)
  - Computational Performance
  - Applications (Computer Vision, Natural Language processing, etc.)
  - Generative AI

- Assessments

No	Assessment Name	Unit Learning Outcomes	Weight %	Individual / Group	Data of release	Due Date and Time
1	Assignment 1	LO1, LO2, LO3, LO6	25%	Individual	Monday 8:00 AM of Study Week 5	Friday 23:59 of Study Week 9
2	Assignment 2	LO1, LO2, LO3, LO4, LO5, LO6	25%	Individual	Monday 8:00 AM of Study Week 11	Friday 23:59 of Study Week 14
3	Final Examination	LO1, LO2, LO3, LO4, LO5	50%	Individual	Specified exam period	Specified exam period

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Labs – What do we need?

- We will mainly use
  - Python as the main programming language
  - Google Colab as an online environment for writing Python Code
    - https://colab.research.google.com/
  - Refer to “Getting Started” section on LMS, which provides details on how to start using Colab

- What kind of exercises?
  - Use existing deep learning libraries (mainly PyTorch) to build and train neural networks and solve some practical problems
  - Implement our own deep learning library (let’s call it MyTorch)
  - Use the library you implemented (MyTorch) to build and train neural networks and solve some practical problems

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In Summary

- Take-home messages
  - Machine Learning can be seen as *programming with data*
  - The focus of this unit is on Deep Learning
  - Start early
  - Do not move to the next topic if you do not understand this week's topic

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This week's lab

- Getting started
  - Familiarize yourself with Python, NumPy and PyTorch
- We have never studied Python – why are we using it in this unit?
  - As an IT graduate, you MUST be able to learn by yourself any programming language
    - ICT159, ICT167, and ICT283 gave you all the necessary foundations and skills you need to learn any programming language
  - You do not need to master it before you get started
    - Learn the basics
    - Learn while your are doing (and delivering)

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Next Week

- Preliminaries
  - Lecture:
    - Warming you up with Linear Neurons
  - Lab
    - Start building up your neurons!

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Questions

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